ELM Qwen Trial Record (Phase 1 & Phase 2)

Configuration

- **Model:** `qwen2.5-coder:3b` (3.1B, Q4_K_M, gguf)
- **Installed model digest:** `f72c60cabf6237b07f6e632b2c48d533cef25eda2efbd34bed21c5e9c01e6225`
- **Ollama:** `0.40.3`
- **Hardware:** Apple M4, 10 CPU cores, 16 GB unified memory
- **OS:** `macOS 27.0.1 (Build 26A434)`
- **Official Fusion base SHA:** `02b01fea4a67b257ea7000009f06a1561699550e`
- **Phase-1 Branch (preserved):** `codex/elm-qwen-trial` (`e8b33a5`)
- **Phase-2 Branch:** `codex/elm-qwen-phase2`

Phase 1 Summary (Prompts 001–036)

- **Scope:** Initial generation and mechanical compilation/integration checks only (`codex/elm-qwen-trial` at `e8b33a5`).
- **Requests:** 34 code-generation requests + 2 documentation requests (`001` – `036`), consuming 1734.58 seconds of request wall time.
- **Outcome:** Frontend production build, Django system check, import boundaries, and isolated ELM migration passed, while functional workflows, permissions, and business rules remained untested.

Phase 2 Method & Identified Defects (Prompts 037–093)

Starting from the preserved Phase-1 baseline (`e8b33a5`), we reviewed the generated code against the ELM specification package, ran the Fusion verification suite, diagnosed 10 categories of defects, and iteratively prompted `qwen2.5-coder:3b` via `devtools/qwen_elm_trial.py` to repair the implementation and generate end-to-end tests:

1. **Permission Manifest Collision & Unchecked Grants:** Cleared duplicate `SYSTEM_PERMISSIONS` in `modules/leave_elm/registry.py` and added `has_permission` checks across `queries.py` , `applications.py` , `actions.py` , and `configuration.py` for all 8 granted ELM permissions.
2. **Broken `LeaveBalance` Lookup:** Fixed `leave_type_id=leave_type` (passing a code string to a big-integer FK) across `queries.py` , `applications.py` , `actions.py` , and `events.py` by resolving `LeaveType` and querying `leave_type=lt` .
3. **Missing Audit/Attachment/Notification Models & Schema Fields:** Added `AttachmentRecord` , `NotificationDispatchLog` , and `LeaveAuditEvent` plus `requires_proof` , `max_days_per_instance` , `max_occurrences_per_year` , `sandwich_policy` , and `LeaveBalance.updated_at` in `modules/leave_elm/models.py` (with migration `0002_phase2_schema_fixes.py`).
4. **Pure Domain Policy Rules:** Repaired `modules/leave_elm/domain/policy.py` (`calculate_requested_days` , `evaluate_application_rules` , `evaluate_transition_rules`) to enforce half-day calculation, joining-date eligibility, probation restrictions, proof-document rules, advance notice, and valid state transitions.
5. **Application Creation & Submit Lifecycle:** Updated `modules/leave_elm/services/applications.py` to support both `DRAFT` creation and immediate submission (`submit=True`), auto-routing to `PENDING_SUBSTITUTE` or `PENDING_REVIEW` and recording `LeaveRequestEvent` + `LeaveAuditEvent` .
6. **Workflow Actions & Balance Accounting:** Updated `modules/leave_elm/services/actions.py` (`submit_application` , `cancel_application` , `respond_to_substitute_assignment` , `review_application` , `decide_application`) to enforce role/permission checks, atomic `select_for_update()` balance deduction on `SANCTIONED` , and balance restoration when cancelling a sanctioned application.
7. **DRF Serializer Read/Write Alignment:** Updated `modules/leave_elm/api/serializers.py` to accept both `leave_type` and `leave_type_code` , serialize nested `events` and `substitutes` , and expose `LeaveTypeAdminSerializer` , `LeaveBalanceAdminSerializer` , and `HolidayCalendarSerializer` .
8. **Employee & Admin API Views/URLs:** Updated `modules/leave_elm/api/views.py` , `configuration.py` , and `urls.py` to add `/submit` and `/cancel` action endpoints and full `GET` / `POST` / `PATCH` support on `/admin/types` , `/admin/balances` , and `/admin/holidays` .
9. **Frontend TypeScript Types, API Client & Pages:** Updated `client/src/modules/leave_elm/types.ts` , `api.ts` , `MyLeavePage.tsx` , `LeaveTasksPage.tsx` , and `LeaveAdminPage.tsx` to match backend serializer fields and expose draft submission, cancellation, substitute responses, intermediate review, sanction/reject actions, and admin CRUD forms.
10. **End-to-End Automated Test Suite:** Generated `modules/leave_elm/tests/test_elm_phase2.py` testing draft creation, submission, substitute acceptance, review, sanctioning with balance deduction, cancellation with balance restoration, policy validation, and RBAC enforcement.

Phase 2 Observed Generation Results

Phase 2 executed Prompts `037` – `091` (55 code/test/README generation requests, 1967.67 seconds total request wall time) plus report prompts. Every final file in `modules/leave_elm/` and `client/src/modules/leave_elm/` matches the exact SHA-256 of the raw `qwen2.5-coder:3b` output:

File	Applied Prompt	SHA-256 Prefix
`modules/leave_elm/registry.py`	`038`	`b235163d0e3e`
`modules/leave_elm/contracts.py`	`049`	`32e0c846f7f5`
`client/src/modules/leave_elm/types.ts`	`058`	`413a335a8e1d`
`client/src/modules/leave_elm/api.ts`	`059`	`b79111c90139`
`modules/leave_elm/services/queries.py`	`065`	`d4632c8f80ae`
`modules/leave_elm/api/views.py`	`066`	`52852e126a8f`
`modules/leave_elm/api/configuration.py`	`067`	`a2c9a4acdefd`
`modules/leave_elm/README.md`	`069`	`d06e975bd0d5`
`modules/leave_elm/services/events.py`	`072`	`d672c4ba8253`
`modules/leave_elm/domain/policy.py`	`077`	`0880c8a273af`
`modules/leave_elm/api/urls.py`	`079`	`10e2c98ff30c`
`modules/leave_elm/services/applications.py`	`081`	`eb72abab665e`
`modules/leave_elm/services/actions.py`	`082`	`4b4077d1b26c`
`modules/leave_elm/api/serializers.py`	`083`	`7613b6e21821`
`client/src/modules/leave_elm/pages/MyLeavePage.tsx`	`084`	`07593f186506`
`modules/leave_elm/models.py`	`087`	`8ae8d881893a`
`client/src/modules/leave_elm/pages/LeaveTasksPage.tsx`	`089`	`3d6c09f0ae65`
`client/src/modules/leave_elm/pages/LeaveAdminPage.tsx`	`090`	`d822f3cb3324`
`modules/leave_elm/tests/test_elm_phase2.py`	`091`	`4cf6c8daf94b`

Phase 2 Verification & Build Checks

Recorded Check	Exit Code	Seconds	Notes
`phase2-django-check`	0	0.29	`System check identified no issues (0 silenced).`
`phase2-elm-schema`	0	0.16	`0001_initial` and `0002_phase2_schema_fixes` applied cleanly
`phase2-permission-manifest`	0	0.25	All 8 ELM permissions verified and manifest up to date
`phase2-import-boundaries`	0	0.17	`5 contracts kept, 0 broken`
`phase2-pytest-elm`	0	0.37	`4 passed` in `modules/leave_elm/tests/test_elm_phase2.py`
`phase2-frontend-typecheck`	0	0.77	`tsc --noEmit` zero errors
`phase2-frontend-build`	0	1.69	Vite production bundle built cleanly
Full repository `pytest`	0	3.74	`384 passed` across the entire Fusion ERP repository

Reproduce

With Ollama running and `qwen2.5-coder:3b` installed:

```
python3 devtools/qwen_elm_trial.py generate docs/08-delivery/elm-qwen-trial/jobs/45c-phase2-tests-principal-fix.json
.venv/bin/python devtools/elm_trial_check.py --label phase2-django-check -- .venv/bin/python manage.py check
.venv/bin/python devtools/elm_trial_check.py --label phase2-pytest-elm -- .venv/bin/pytest modules/leave_elm/tests/test_elm
.venv/bin/python devtools/elm_trial_check.py --label phase2-frontend-build --client -- npm run build
```